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https://keshavbhandari.github.io

Keshav Bhandari

Summary

I am a PhD student at the Centre for Digital Music (C4DM) at Queen Mary University of London. My research revolves around neuro-symbolic approaches for generating music with a focus on controllability and musical structure.

Education

Queen Mary University of London, EECS — London, UK
Ph.D. in Artificial Intelligence — 2023 - 2027 *(expected)*
- Advisors: Simon Colton; Geraint Wiggins
- Research area: Neuro Symbolic Automated Music Composition

Northwestern University, McCormick School of Engineering — Evanston, USA
Master of Science in Artificial Intelligence (GPA: 3.96/4.0) — 2021 - 2022

Purdue University, Krannert School of Management — Indiana, USA
Bachelor of Science in Business Management & Marketing — 2012 - 2016
- Dean's List, 2013

Publications

1. **Bhandari, K.**, Bizzarri, M., Wiggins, G. A., & Colton, S. (2026). Change is Key: A Generative Framework for Controllable Musical Modulations. Manuscript under review, JNMR.
2. **Bhandari, K.**, Chang, S., Roy, A., Ronchini, F., Benetos, E., Herremans, D., & Colton, S. (2026). Text2Score: Generating Sheet Music From Textual Prompts. Manuscript under review, AAAI.
3. **Bhandari, K.**, Roy, A., Herremans, D., & Colton, S. Emerging AI Technologies for Music: Towards Controllable, Collaborative, and Creative Systems. *Proceedings of Machine Learning Research (PMLR)*, vol. 303, pp. 1–5, 1st Workshop on Emerging AI Technologies for Music (EAIM) at AAAI, 2026.
4. **Bhandari, K.**, Chang, S., Lu, T., Enus, F. R., Herremans, D., & Colton, S. ImprovNet - Generating Controllable Musical Improvisations with Iterative Corruption Refinement. *International Joint Conference on Neural Networks (IJCNN)*, 2025.
5. **Bhandari, K.**, Wiggins, G. A., & Colton, S. Yin-Yang: Developing Motifs With Long-Term Structure And Controllability. *International Conference on Artificial Intelligence in Music, Sound, Art and Design (EvoMUSART)*, 2025. ***(Best Paper Award)***
6. **Bhandari, K.**, Roy, A., Wang, K., Puri, G., Colton, S., & Herremans, D. Text2midi: Generating Symbolic Music from Captions. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
7. Colton, S., Bradshaw, L., Banar, B., & **Bhandari, K.** Automatic Generation of Expressive Piano Miniatures. *International Conference on Computational Creativity (ICCC)*, 2024.
8. **Bhandari, K.**, & Colton, S. Motifs, Phrases, and Beyond: The Modelling of Structure in Symbolic Music Generation. *In International Conference on Artificial Intelligence in Music, Sound, Art and Design (EvoMUSART)*, 2024.
9. O'Reilly, P., Bugler, A., **Bhandari, K.**, Morrison, M., & Pardo, B. VoiceBlock: Privacy through Real-Time Adversarial Attacks with Audio-to-Audio Models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Professional Experience

Singapore University of Technology and Design | Singapore — Aug 2025 - Nov 2025

- **Research Intern, AMAAI Lab (Advisor: Prof. Dorien Herremans)**
  - Developed **ImprovNet 2**, a framework for genre-controllable musical improvisation using LLaDA (discrete diffusion models).
  - Built **Text-to-Sheet-Music**, a hierarchical model orchestrated by LLMs in a two-step decoding process, enabling text-conditioned music generation without explicit text–music annotations during training.

Boston Consulting Group | Los Angeles, CA, USA — Jan 2023 - Sep 2023

- **Data Scientist**
  - Vodafone Italy: Developed AI-based tools using Llama and Stable Diffusion to enhance digital marketing efficiency.
  - Staples: Designed a pattern matching algorithm to improve control group selection for precise reporting and strategic insights.
  - Walgreens: Built model drift scripts to rectify deviations in price optimization algorithm in a MLOps framework.

Epsilon | Bangalore, Karnataka, India — Sep 2017 - Feb 2021

- **Senior Data Scientist**
  - Built deep learning based personalized recommendation engines using Transformers, boosting client hit rates by up to **5%**.
  - Deployed end-to-end ML & DL solutions as a product on cloud platforms, achieving lift rates between **3X** to **9X**.
  - Developed **Gredel**, an interactive R-Shiny package for automating predictive modeling and improving ML interpretability.

Awards and Honors

- EvoSTAR 2025 Best Paper Award / Outstanding Student Award — 2025
- PhD Studentship - UK Research and Innovation (UKRI) — 2023
- 2nd place in Epsilon India Recommendation Engine Hackathon (26 Teams) — 2020
- Top 6% Kaggle Categorical Feature Encoding Challenge (1,342 Teams) — 2019
- Champion Innovator Spot Award, Epsilon — 2019
- Top 3% Kaggle TalkingData Ad-Tracking Fraud Detection Challenge (3,946 Teams) — 2018

Grants Awarded

"ImprovNet2: Generating Musical Improvisations with Discrete Diffusion"
EuroHPC Development Access Compute Grant. 3,500 GPU hours on the JUPITER Booster partition. Proposal ID: EHPC-DEV-2026D09-047. — Sep 2026

- PIs: Simon Colton and Emmanouil Benetos
- Role: I wrote this grant proposal in full, based on my ImprovNet2 research project.

"Using deep learning to build a robust automated recognition system for rare and endangered birds in India"
AI for Earth Microsoft Azure Research Credits Grant. $5,000. — Oct 2021

- PI: Keshav Bhandari
- Role: I substantially wrote and edited this grant, and it was based on my work.

Teaching

- **Guest Lecturer & Demonstrator**, Computational Creativity — Jan 2025 – May 2025

  5 Lecture Topics: Generative AI for Creative Language, Generative Music and Audio, Evolutionary Computation, and Evaluating Generative Systems.

- **Senior Demonstrator**, Data Science and AI master's cohort — Sep 2024 - April 2025
- **Teaching Fellow**, Supervised 6 UG and 12 PG students on FYP — Oct 2024 - July 2025
- **Demonstrator**, Computational Creativity — Jan 2025 - May 2025
- **Demonstrator**, Statistics for AI and Data Science — Sep 2024 - Dec 2024
- **PhD Mentor**, Google DeepMind Research Ready Programme — Jun 2024 - Jul 2024

Invited Talks & PresentATIONS

- **Invited Talk**: *ImprovNet - Generating Controllable Musical Improvisations with Iterative Corruption Refinement*. Northwestern University, McCormick School of Engineering. Feb. 13th, 2025.
- **Guest Lecture**: *Limitations of Sequential Models and An Introduction To Transformers*. Queen Mary University of London, Google Deepmind Research Ready Programme Orientation. June 26th, 2025.
- **Guest Lecture**: *Self-Supervised Corruption-Refinement Models for Low Resource Generative Music*. Queen Mary University of London, Class: Computational Creativity. April 2nd, 2025.
- **Guest Lecture**: *How To Be An Expert At Literature Reviews*. Queen Mary University of London, Google Deepmind Research Ready Programme Orientation. June 28th, 2024.
- **Research Presentation**: *Towards Melodic Development with Discrete Diffusion Models for Symbolic Music*. Queen Mary University of London, Digital Music Research Network One-Day Workshop (DMRN+18). December 2023.
- **Research Presentation**: *Augmentations to improve rare bird call classification for a highly imbalanced multi-label soundscape environment*. Digital University Kerala, 11th International Conference on Ecological Informatics (ICEI 2020+1). November 2021.

Academic & Community Service

Conference Volunteer, International Society for Music Information Retrieval (ISMIR) — 2026

Reviewer, Journal of Creative Music Systems — 2026

Workshop Chair | Lead Organizer, 1st International Workshop on Emerging AI Technologies for Music (EAIM-26), AAAI-26 — 2026

Session Chair, Special Track: Human-AI Interaction in Creative Arts and Sciences, IJCNN — 2025

Reviewer, IETE Technical Review — 2026

Reviewer, The Association for the Advancement of Artificial Intelligence (AAAI) — 2025

Reviewer, International Joint Conference on Neural Networks (IJCNN) — 2025

Conference Volunteer, Association for the Advancement of Artificial Intelligence (AAAI) — 2024

Member, Data Kind, Bangalore Chapter — Aug 2018–Oct 2019

President, Purdue Marketing Association, Purdue University — Jan 2015–Dec 2015